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Chapter 1

Introduction

Hand exoskeletons have gained increasing attention in recent years due to their potential in rehabilitation, assistive technology, and human motion augmentation. By providing controlled support to the hand and fingers, such systems may help restore function in individuals with impaired motor ability, while also serving as platforms for studying human biomechanics and control. Because the hand is mechanically complex and highly dexterous, the development of effective hand exoskeletons requires accurate models of finger motion and dynamics.

A key challenge in the design of such systems is the interaction between the mechanical structure of the exoskeleton and the physiological properties of the human finger. The finger is a multi-joint system with coupled motion, passive elastic properties, and nonlinear dynamics. To design controllers that achieve stable and precise movement assistance, it is therefore important to establish a mathematical model that captures the essential kinematic and dynamic behavior of the finger, while remaining suitable for simulation and control design.

Chapter 2

Theory

2.1 Anatomy

The hand is a complex biomechanical system, comprised of many different bones, muscles, tendons, and ligaments. To explain different aspects of the hand in relation to one another, we use *anatomical terminology* [National Cancer Institute, 2026], or more specifically *anatomical directional terms*. Additionally when describing actions of muscles upon the skeleton, we use *anatomical terms of movement* [TeachMeAnatomy, 2026b].

2.1.1 Anatomical directional terms

Anatomical directional terms are used in relation to the *anatomical position*, shown in Figure 2.1, where the body is assumed standing upright, with arms at the sides and palms facing forward. Relevant terms include: *proximal*, *distal*.

- **Proximal** means toward or nearest the point of origin of a part.
- **Distal** means away from or farthest from the point of origin of a part.

2.1.2 Anatomical terms of movement

To explain the actions of muscles upon the skeleton, we use anatomical terms of movement, which describes the movement of a body part in relation to the anatomical position. Relevant terms include: *flexion*, *extension*, *abduction*, *adduction*.

- **Flexion** is a movement that decreases the angle between two body parts.
- **Extension** is a movement that increases the angle between two body parts.
- **Abduction** is a movement away from the midline of the body.

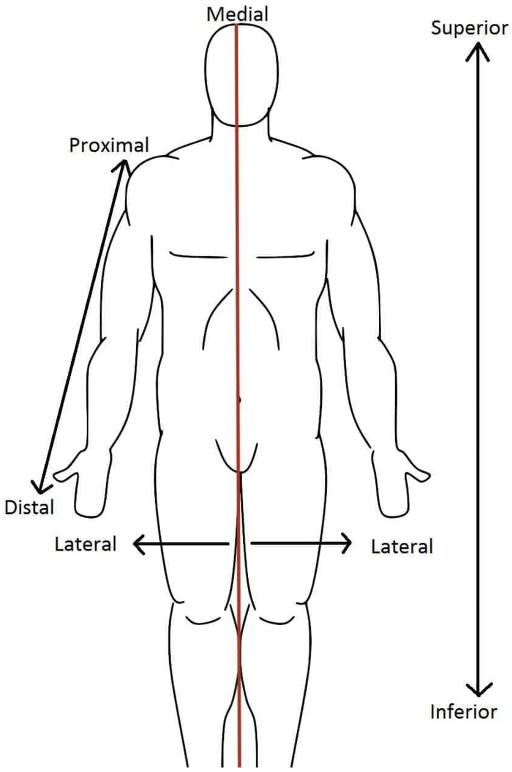

Figure 2.1: The anatomical position, with the corresponding directional terms. Image from TeachMeAnatomy, 2026a.

- **Adduction** is a movement toward the midline of the body.

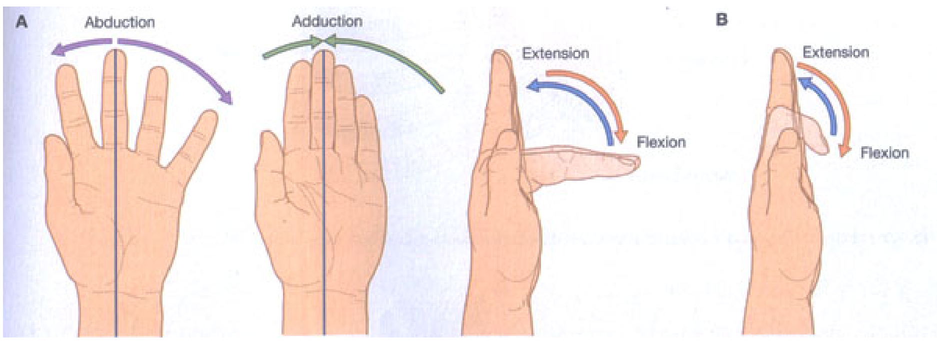

Figure 2.2: Anatomical terms of movement. Image from TeachMeAnatomy, 2026b.

2.1.3 Hand anatomy

The hand is comprised of several bones, muscles, tendons, and ligaments. The bones, illustrated in Figure 2.3, include the *carpals*, *metacarpals*, and *phalanges*. The carpals make out the wrist, the metacarpals make out the palm, and the phalanges make out the fingers. The thumb has two phalanges, while the other fingers have three. The bones of interest are the phalanges (marked in green), as they are the ones being moved by the exoskeleton. The metacarpals are assumed stationary for the task at hand, as the exoskeleton will provide stiffness to the palm and wrist.

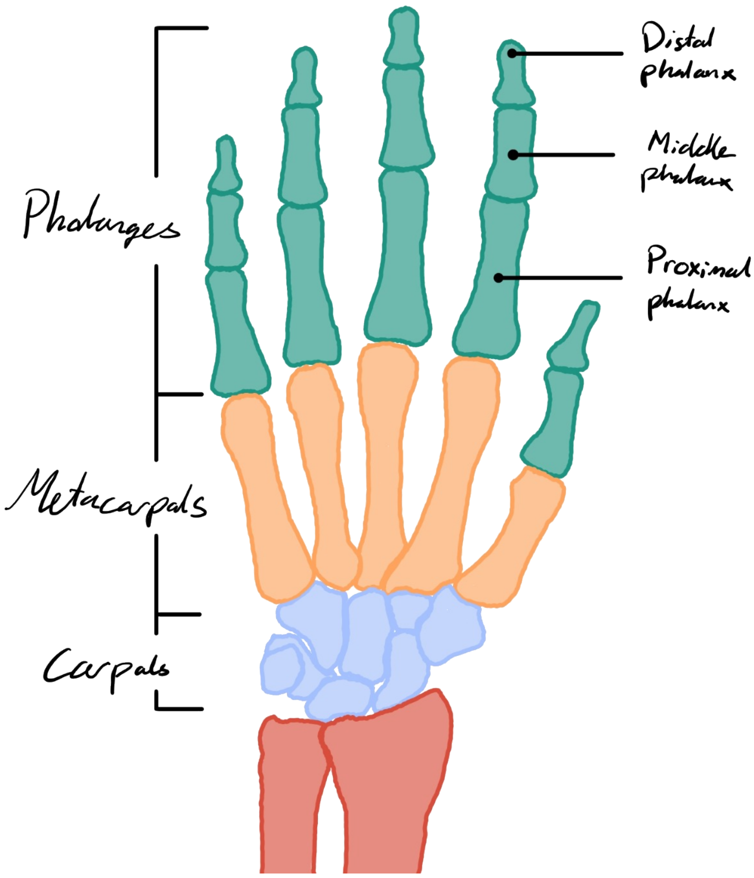

Figure 2.3: The bones of the hand. Image from author.

The phalanges are further divided into three sections: the *proximal phalanges*, the *middle phalanges*, and the *distal phalanges*. The proximal phalanges are the ones closest to the palm, while the distal phalanges are the ones furthest from the palm. The middle phalanges are the ones in between. The thumb does not have a middle phalanx, as it only has two phalanges.

Further details about the joints are shown in Figure 2.4. The joints of interest are the *metacarpophalangeal joints* (MCP), the *proximal interphalangeal joints* (PIP), and the *distal interphalangeal joints* (DIP). The MCP joints are the ones between the metacarpals and the proximal phalanges, while the PIP joints are the ones between the proximal phalanges and the middle phalanges, and the DIP joints are the ones between the middle phalanges and the distal phalanges. The thumb only has an MCP joint and an interphalangeal joint (IP), as it does not have a middle phalanx.

At each of the joints, there are collateral ligaments that provide stability. For the PIP, IP and DIP joints, they provide stiffness that prevents side to side motion. For the MCP joints they're relaxed when the finger is extended, facilitating for adduction and abduction, but stiff when the finger is flexed.

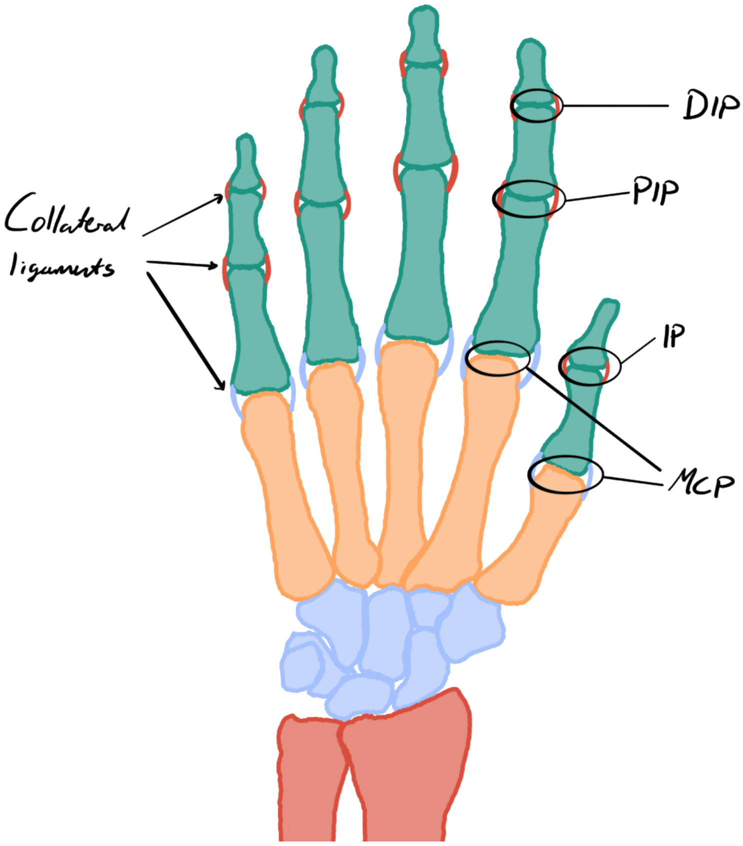

Figure 2.4: The joints of the hand. Image from author.

2.2 Modelling

2.2.1 Homogenous transformations

When modelling kinematic chains, we often need to move between frames. This often includes both rotation and translation. An efficient way to represent this is by using *homogenous transformations*.

Definition

For a 2D system, the rotation matrix for a counter-clockwise rotation by an angle θ is given by:

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}. \tag{2.1}$$

Given a translation vector from frame i to frame j, written in terms of frame i, $\mathbf{d}_i^j = \begin{bmatrix} d_x^i \\ d_y^i \end{bmatrix}$, the homogenous transformation matrix from frame j to frame i, describing the rotation θ and translation $\mathbf{d}_i^j$ from frame i to frame j, can be expressed as:

$$\mathbf{H}_j^i = \begin{bmatrix} \mathbf{R}(\theta) & \mathbf{d}_i^j \\ \mathbf{0}_{1\times 2} & 1 \end{bmatrix}. \tag{2.2}$$

Points represented in terms of homogenous transformations are represented as a 3D vector, where the first two elements represent the coordinates of the point in the plane, and the third element is always 1. For example, a point $\mathbf{p}_n$ in frame n with coordinates (x, y) can be represented as:

$$\mathbf{p}_n = \begin{bmatrix} p_x^n \\ p_y^n \\ 1 \end{bmatrix}. \tag{2.3}$$

Useful properties

Given a kinematic chain of size n, we can express the homogenous transformation from frame n to frame 1 as the product of the homogenous transformations between each consecutive frame:

$$\mathbf{H}_n^1 = \mathbf{H}_2^1 \mathbf{H}_3^2 \cdots \mathbf{H}_n^{n-1}. \tag{2.4}$$

Given a point $\mathbf{p}_n$ in frame n, we can express it in frame 1 as:

$$\mathbf{p}_1 = \begin{bmatrix} p_x^1 \\ p_y^1 \\ 1 \end{bmatrix} = \mathbf{H}_2^1\mathbf{H}_3^2 \cdots \mathbf{H}_n^{n-1}\mathbf{p}_n = \mathbf{H}_n^1\mathbf{p}_n. \tag{2.5}$$

2.3 Control

2.3.1 State-Space representation

A way to express a n'th order ordinary differential equation (ODE) is to express it as a system of n first order ODEs. This is called the *state-space representation* of the system. A state-space representation commonly used in control theory for linear time-invariant (LTI) systems is the following:

$$\begin{aligned} &\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) + \mathbf{d}(t), \quad && \mathbf{x}(t) \in \mathbb{R}^n, \quad \mathbf{u}(t) \in \mathbb{R}^m, \quad \mathbf{y}(t) \in \mathbb{R}^p, \quad \mathbf{d}(t) \in \mathbb{R}^n, \\ &\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t), && \mathbf{A} \in \mathbb{R}^{n\times n}, \quad \mathbf{B} \in \mathbb{R}^{n\times m}, \quad \mathbf{C} \in \mathbb{R}^{p\times n}. \end{aligned} \tag{2.6}$$

Here, $\mathbf{x}(t)$ is the state vector, which contains all the information about the system at time t. $\mathbf{u}(t)$ is the input vector at time t. $\mathbf{y}(t)$ is the output vector, which contains all the measurements of the system at time t. $\mathbf{d}(t)$ is a disturbance vector.

2.3.2 Controllability

A system is said to be *controllable* if it is possible to drive the state of the system from any initial state to any desired final state in finite time, using an appropriate control input. For a LTI system, controllability can be determined by the rank of the controllability matrix $\mathcal{C}$, defined as:

$$\mathcal{C} = [\mathbf{B}, \mathbf{AB}, \mathbf{A}^2\mathbf{B}, \ldots, \mathbf{A}^{n-1}\mathbf{B}]. \tag{2.7}$$

If $\text{rank}(\mathcal{C}) = n$, where n is the dimension of the state vector $\mathbf{x}$, then the system is controllable.

2.3.3 MRAC

Model Reference Adaptive Control (MRAC) is a control strategy that aims to make the output of a system follow the output of a reference model, even in the presence of uncertainties or disturbances. The following definition of MRAC is based on [**Cite Robust Adaptive Control book, 6.2.3**].

Given the nth order LTI-system in state-space form,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \quad \mathbf{x}(t) \in \mathbb{R}^n, \mathbf{u}(t) \in \mathbb{R}^m, \tag{2.8}$$

and the reference model,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}_m(t) = \mathbf{A}_m\mathbf{x}_m(t) + \mathbf{B}_m\mathbf{r}(t), \quad \mathbf{x}_m(t) \in \mathbb{R}^n, \mathbf{r}(t) \in \mathbb{R}^m. \tag{2.9}$$

The reference model and input r(t) are chosen so that $\mathbf{x}_m(t)$ represents the desired trajectory that $\mathbf{x}(t)$ should follow. Crucially, $\mathbf{A}_m$ must be a stable matrix for this to work. If matrices $\mathbf{A}$ and $\mathbf{B}$ are known, then the control input $\mathbf{u}(t)$ can be designed as follows:

$$\mathbf{u}(t) = -\mathbf{K}^*\mathbf{x}(t) + \mathbf{L}^*\mathbf{r}(t), \quad \mathbf{K}^* \in \mathbb{R}^{m\times n}, \mathbf{L}^* \in \mathbb{R}^{m\times m}. \tag{2.10}$$

This obtains the closed loop system:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}(t) = (\mathbf{A} - \mathbf{B}\mathbf{K}^*)\mathbf{x}(t) + \mathbf{B}\mathbf{L}^*\mathbf{r}(t). \tag{2.11}$$

If we choose $\mathbf{K}^*$ and $\mathbf{L}^*$ such that,

$$\mathbf{A} - \mathbf{B}\mathbf{K}^* = \mathbf{A}_m, \qquad \mathbf{B}\mathbf{L}^* = \mathbf{B}_m, \tag{2.12}$$

then the closed loop system becomes:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}(t) = \mathbf{A}_m\mathbf{x}(t) + \mathbf{B}_m\mathbf{r}(t). \tag{2.13}$$

Hence the closed-loop dynamics of $\mathbf{x}(t)$ is the same to Equation (2.9), and $\mathbf{x}(t) \to \mathbf{x}_m(t)$ exponentially fast. But for most cases, $\mathbf{A}$ and $\mathbf{B}$ are not known, which means that $\mathbf{K}^*$ and $\mathbf{L}^*$ has to be estimated online. Assuming that $\mathbf{K}^*$ and $\mathbf{L}^*$ exist, we'll change the input in Equation (2.10) to:

$$\mathbf{u}(t) = -\mathbf{K}(t)\mathbf{x}(t) + \mathbf{L}(t)\mathbf{r}(t), \quad \mathbf{K}(t) \in \mathbb{R}^{m\times n}, \mathbf{L}(t) \in \mathbb{R}^{m\times m}. \tag{2.14}$$

Adaptive Law

Defining the tracking error $\mathbf{e}(t)$ as:

$$\mathbf{e}(t) = \mathbf{x}(t) - \mathbf{x}_m(t). \tag{2.15}$$

Solving for $\mathbf{P}$ for a $\mathbf{Q} = \mathbf{Q}^T > 0$ in the Lyapunov equation:

$$\mathbf{A}_m^T\mathbf{P} + \mathbf{P}\mathbf{A}_m = -\mathbf{Q}, \tag{2.16}$$

we have the adaptive law:

$$
\begin{aligned}
\dot{\mathbf{K}}(t) &= \mathbf{B}_m^T \mathbf{P} \mathbf{e} \mathbf{x}^T sgn(l), \\
\dot{\mathbf{L}}(t) &= -\mathbf{B}_m^T \mathbf{P} \mathbf{e} \mathbf{r}^T sgn(l).
\end{aligned}
\tag{2.17}
$$

Chapter 3

Modelling

3.1 Finger model

For modelling the finger, we're assuming that the motion is limited to a 2D-plane. This is mostly true, due to the collateral ligaments at the DIP- and the PIP-joint, preventing excessive side-to-side bending, as discussed in [**Reference to theory:planar manipulator**]. The MCP joint doesn't have the same support-structure, so it's free to move side to side, but this isn't as relevant to model for the task at hand, as the lateral movement from the cable pulling on the finger is minimal.

For our model, we assume that the finger is composed of three links, with three revolute joints, like a *planar manipulator* (**Reference to theory:planar manipulator**). To model the stiffness from the tendons and ligaments, we'll add a torsional spring at each joint. The springs will have a non-zero equilibrium angle, representing the natural resting position of the finger joints. The damping from the soft tissue will be modeled by torsional dampers. A schematic of the model can be seen in Figure 3.1. We'll model the links as rigid cylinders, with uniform mass distribution, and the joints as ideal revolute joints, with no friction. The parameters of the model are listed in Figure 3.2.

To model the dynamics of the finger, we'll use the Euler-Lagrange equations as shown in (**Add reference to theory: Euler-Lagrange Equations**). We'll define our generalized coordinates as the relative joint angles, φ_{MCP}, φ_{PIP} and φ_{DIP}:

$$\mathbf{q} \triangleq \begin{bmatrix} \varphi_{\text{MCP}} \\ \varphi_{\text{PIP}} \\ \varphi_{\text{DIP}} \end{bmatrix} \implies \dot{\mathbf{q}} = \begin{bmatrix} \dot{\varphi}_{\text{MCP}} \\ \dot{\varphi}_{\text{PIP}} \\ \dot{\varphi}_{\text{DIP}} \end{bmatrix} \implies \ddot{\mathbf{q}} = \begin{bmatrix} \ddot{\varphi}_{\text{MCP}} \\ \ddot{\varphi}_{\text{PIP}} \\ \ddot{\varphi}_{\text{DIP}} \end{bmatrix}. \tag{3.1}$$

To derive the equations of motion, we need to define the kinetic energy, T, and the potential energy, V of our system.

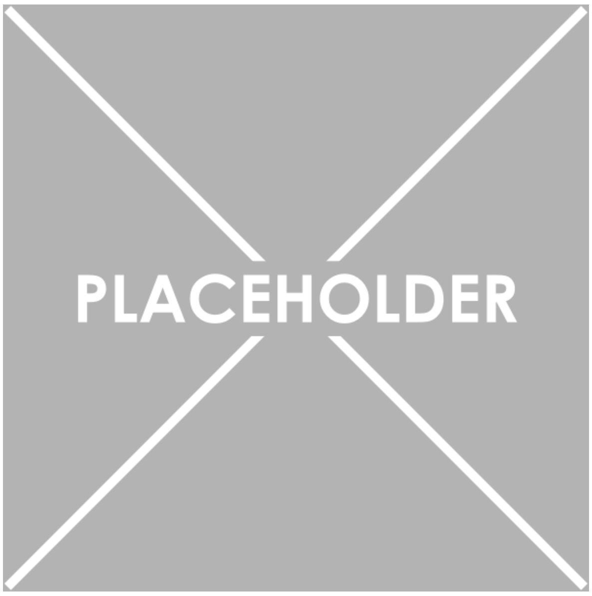

Figure 3.1: Schematic of the finger model, with three revolute joints, torsional springs and dampers.

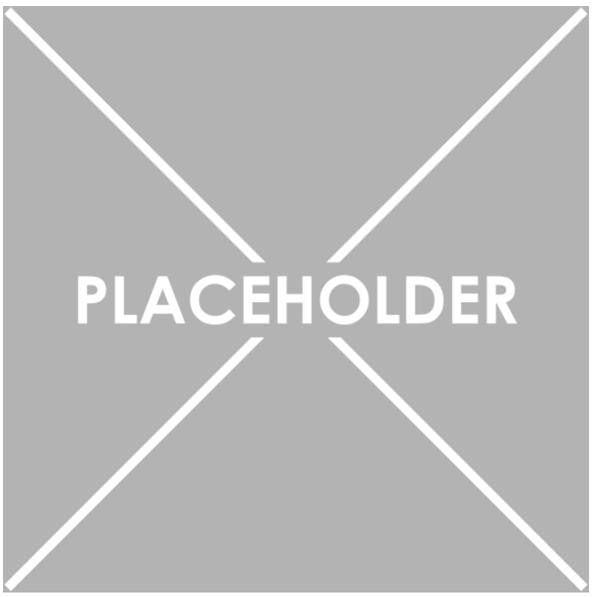

Figure 3.2: Table of model parameters.

3.1.1 Kinetic energy

The kinetic energy will be the sum of the kinetic energy of each link, which can be decomposed into the rotational and translational kinetic energy, defined as:

$$T_{\text{tot}} = T_{\text{rot}} + T_{\text{trans}}. \tag{3.2}$$

Rotational kinetic energy

The rotational kinetic energy of the system is rather straightforward to derive, as the angular velocity of each link is equal to the time derivative of the joint angle. Since the angles are relative, the total angular velocity of each link is the sum of the time derivatives of the joint angles up to that link. Hence, the rotational kinetic energy can be expressed as:

$$T_{\text{rot}} = \frac{1}{2}J_{\text{MCP}}\dot{\varphi}_{\text{MCP}}^2 + \frac{1}{2}J_{\text{PIP}}(\dot{\varphi}_{\text{MCP}} + \dot{\varphi}_{\text{PIP}})^2 + \frac{1}{2}J_{\text{DIP}}(\dot{\varphi}_{\text{MCP}} + \dot{\varphi}_{\text{PIP}} + \dot{\varphi}_{\text{DIP}})^2. \tag{3.3}$$

Translational kinetic energy

The translational kinetic energy is a bit more complex to derive, as the linear velocity of each link depends on the geometry of the system. The linear velocity of each link can be derived by finding the position of the center of mass of each link and taking the time derivative. To find the position of the center of mass, we'll use homogenous transformations, as described in Section 2.2.1.

Let's start by defining frames for moving between the links. We'll define the static world frame as frame i, attached to the MCP joint, where the x-axis is alligned with the metacarpal. The next is frame PP, attached to the PIP joint, where the x-axis is alligned with the proximal phalanx. Subsequently we have frame MP, attached to the DIP joint, where the x-axis is alligned with the middle phalanx. Finally, we have frame DP, attached to the tip of the finger, where the x-axis is alligned with the distal phalanx. The frames are shown in Figure 3.3. The homogenous transformations between these frames can be expressed as:

$$\begin{aligned}
\mathbf{H}_{PP}^{i} &= \begin{bmatrix} \mathbf{R}(\varphi_{\text{MCP}}) & \mathbf{R}(\varphi_{\text{MCP}})\mathbf{e}_1 l_{\text{PP}} \\ \mathbf{0}_{1\times 2} & 1 \end{bmatrix}, \\
\mathbf{H}_{MP}^{PP} &= \begin{bmatrix} \mathbf{R}(\varphi_{\text{PIP}}) & \mathbf{R}(\varphi_{\text{PIP}})\mathbf{e}_1 l_{\text{MP}} \\ \mathbf{0}_{1\times 2} & 1 \end{bmatrix}, \\
\mathbf{H}_{DP}^{MP} &= \begin{bmatrix} \mathbf{R}(\varphi_{\text{DIP}}) & \mathbf{R}(\varphi_{\text{DIP}})\mathbf{e}_1 l_{\text{DP}} \\ \mathbf{0}_{1\times 2} & 1 \end{bmatrix}.
\end{aligned} \tag{3.4}$$

Figure 3.3: Schematic of the homogenous transformation frames.

Assuming the the center of mass of each link is located at the midpoint of the link, we can express the position of the center of mass of each link in the world frame as:

$$\begin{aligned}
\mathbf{p}^i_{\text{PP,CM}} &= \mathbf{H}^i_{PP} \begin{bmatrix} -l_{\text{PP}}/2 \\ 0 \\ 1 \end{bmatrix}, \\
\mathbf{p}^i_{\text{MP,CM}} &= \mathbf{H}^i_{PP}\mathbf{H}^{PP}_{MP} \begin{bmatrix} -l_{\text{MP}}/2 \\ 0 \\ 1 \end{bmatrix}, \\
\mathbf{p}^i_{\text{DP,CM}} &= \mathbf{H}^i_{PP}\mathbf{H}^{PP}_{MP}\mathbf{H}^{MP}_{DP} \begin{bmatrix} -l_{\text{DP}}/2 \\ 0 \\ 1 \end{bmatrix}.
\end{aligned} \tag{3.5}$$

Extracting the x and y components of the position, defining them as $\mathbf{r}_{\text{PP,CM}}$, $\mathbf{r}_{\text{MP,CM}}$, and $\mathbf{r}_{\text{DP,CM}}$, we can take the time derivative to find the linear velocity of each center of mass, which can then be used to calculate the translational kinetic energy as:

$$T_{\text{trans}} = \frac{1}{2}\left[m_{\text{PP}}\dot{\mathbf{r}}^T_{\text{PP,CM}}\dot{\mathbf{r}}_{\text{PP,CM}} + m_{\text{MP}}\dot{\mathbf{r}}^T_{\text{MP,CM}}\dot{\mathbf{r}}_{\text{MP,CM}} + m_{\text{DP}}\dot{\mathbf{r}}^T_{\text{DP,CM}}\dot{\mathbf{r}}_{\text{DP,CM}}\right]. \tag{3.6}$$

3.1.2 Potential energy

For the potential energy one would normally consider the gravitational potential energy. However, since the finger is relatively light and the motion during gripping tasks is mostly in the horizontal plane, we'll omit the gravitational potential energy for simplicity. Also, for clarity we'll model the torsional springs as generalized forces, instead of including them in the potential energy. Hence, the potential energy of the system can be expressed as:

$$V = 0. \tag{3.7}$$

3.1.3 Generalized forces

The generalized forces in our system will be the torques from the torsional springs and dampers, aswell as the torques from the cable pulling on the finger, and other external torques when the finger interacts with objects. The torques from the torsional springs can be expressed as:

$$\begin{gathered}
\boldsymbol{\tau}_{\text{spring}} = \begin{bmatrix} k_{\text{MCP}}(\varphi_{\text{MCP}} - \varphi_{\text{MCP,eq}}) \\ k_{\text{PIP}}(\varphi_{\text{PIP}} - \varphi_{\text{PIP,eq}}) \\ k_{\text{DIP}}(\varphi_{\text{DIP}} - \varphi_{\text{DIP,eq}}) \end{bmatrix} = \mathbf{K}\left(\mathbf{q} - \mathbf{q}_{\text{eq}}\right) \\
\mathbf{K} \triangleq \begin{bmatrix} k_{\text{MCP}} & 0 & 0 \\ 0 & k_{\text{PIP}} & 0 \\ 0 & 0 & k_{\text{DIP}} \end{bmatrix}, \quad \mathbf{q}_{\text{eq}} \triangleq \begin{bmatrix} \varphi_{\text{MCP,eq}} \\ \varphi_{\text{PIP,eq}} \\ \varphi_{\text{DIP,eq}} \end{bmatrix}.
\end{gathered} \tag{3.8}$$

The torques from the torsional dampers can be expressed as:

$$\boldsymbol{\tau}_{\text{damper}} = \begin{bmatrix} b_{\text{MCP}}\dot{\varphi}_{\text{MCP}} \\ b_{\text{PIP}}\dot{\varphi}_{\text{PIP}} \\ b_{\text{DIP}}\dot{\varphi}_{\text{DIP}} \end{bmatrix} = \mathbf{B}\dot{\mathbf{q}}, \quad \mathbf{B} \triangleq \begin{bmatrix} b_{\text{MCP}} & 0 & 0 \\ 0 & b_{\text{PIP}} & 0 \\ 0 & 0 & b_{\text{DIP}} \end{bmatrix}. \tag{3.9}$$

The total generalized forces can then be expressed as:

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{\text{ext}} + \boldsymbol{\tau}_{\text{cable}} - \boldsymbol{\tau}_{\text{spring}} - \boldsymbol{\tau}_{\text{damper}}. \tag{3.10}$$

3.1.4 Rigid-body dynamics

For rigid-body mechanical systems, like the one we're modelling, the Euler-Lagrange equations can be expressed in matrix form as (**Add reference to theory: Rigid-body dynamics**):

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \boldsymbol{\tau}. \tag{3.11}$$

The *inertia matrix* $\mathbf{M}(\mathbf{q})$ is a symmetric positive-definite matrix that represents the mass distribution of the system. We'll define $\varphi_{\text{MCP}} \triangleq \varphi_1$, $\varphi_{\text{PIP}} \triangleq \varphi_2$, and $\varphi_{\text{DIP}} \triangleq \varphi_3$, the cumulative angles $\varphi_{12} \triangleq \varphi_1 + \varphi_2$, $\varphi_{23} \triangleq \varphi_2 + \varphi_3$, and the shorthand form $c_\varphi \triangleq \cos(\varphi)$, $s_\varphi \triangleq \sin(\varphi)$. Based on the E.L.-equations the inertia matrix in our system then equates to:

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} p_1 + p_2 c_{\varphi_2} + p_3 c_{\varphi_{23}} + p_4 c_{\varphi_3} & p_4 c_{\varphi_3} + p_5 + p_6 c_{\varphi_2} + p_7 c_{\varphi_{23}} & p_7 c_{\varphi_{23}} + p_8 + p_9 c_{\varphi_3} \\ p_4 c_{\varphi_3} + p_5 + p_6 c_{\varphi_2} + p_7 c_{\varphi_{23}} & p_4 c_{\varphi_3} + p_5 & p_8 + p_9 c_{\varphi_3} \\ p_7 c_{\varphi_{23}} + p_8 + p_9 c_{\varphi_3} & p_8 + p_9 c_{\varphi_3} & p_8 \end{bmatrix}, \tag{3.12}$$

where the different p_i's are shown in Table 3.1. The *coriolis and centrifugal matrix* $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ represents the coriolis and centrifugal forces that arise from the motion of the system. For our system, it equates to:

$$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{2} \begin{bmatrix} \gamma_{11} & \gamma_{12} & \gamma_{13} \\ \gamma_{21} & \gamma_{22} & \gamma_{23} \\ \gamma_{31} & \gamma_{32} & 0 \end{bmatrix}, \tag{3.13}$$

where

$$\begin{aligned}
\gamma_{11} &= -p_2 \dot{\varphi}_2 s_{\varphi_2} - p_3 \dot{\varphi}_{2-3} s_{\varphi_{23}} - p_4 \dot{\varphi}_3 s_{\varphi_3}, \\
\gamma_{12} &= -\left(p_2 \dot{\varphi}_1 + 2 p_6 \dot{\varphi}_2\right) s_{\varphi_2} - \left(p_3 \dot{\varphi}_1 + 2 p_7 \dot{\varphi}_{2-3}\right) s_{\varphi_{23}} - p_4 \dot{\varphi}_3 s_{\varphi_3}, \\
\gamma_{13} &= -\left(p_3 \dot{\varphi}_1 + 2 p_7 \dot{\varphi}_{2-3}\right) s_{\varphi_{23}} - \left(p_4 \dot{\varphi}_{1-2} + 2 p_9 \dot{\varphi}_3\right) s_{\varphi_3}, \\
\gamma_{21} &= \dot{\varphi}_1 \left(p_2 s_{\varphi_2} + p_3 s_{\varphi_{23}}\right) - \dot{\varphi}_3 p_4 s_{\varphi_3}, \\
\gamma_{22} &= -\dot{\varphi}_3 p_4 s_{\varphi_3}, \\
\gamma_{23} &= -\left(p_4 \dot{\varphi}_{1-2} + 2 p_9 \dot{\varphi}_3\right) s_{\varphi_3}, \\
\gamma_{31} &= \dot{\varphi}_1 p_3 s_{\varphi_{23}} + p_4 \dot{\varphi}_{1-2} s_{\varphi_3}, \\
\gamma_{32} &= p_4 \dot{\varphi}_{1-2} s_{\varphi_3}.
\end{aligned}$$

The generalized forces $\boldsymbol{\tau}$ are the same as in Equation (3.10). The *gravity vector* $\mathbf{G}(\mathbf{q})$ represents the gravitational forces acting on the system. Since we're omiting the effects of gravity, the gravity vector is zero:

$$\mathbf{G}(\mathbf{q}) = \mathbf{0}. \tag{3.14}$$

Inserting all the results above, and solving for $\ddot{\mathbf{q}}$, we can express the equations of motion of our system as:

$$\ddot{\mathbf{q}} = \mathbf{M}(\mathbf{q})^{-1} \left[\boldsymbol{\tau}_{\text{ext}} + \boldsymbol{\tau}_{\text{cable}} - \left(\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{B}\right) \dot{\mathbf{q}} - \mathbf{K}\left(\mathbf{q} - \mathbf{q}_{\text{eq}}\right)\right]. \tag{3.15}$$

Table 3.1: Constants in system matrices.

Constant	Value
p_1	$J_{\text{MCP}} + J_{\text{PIP}} + J_{\text{DIP}} + \frac{l_{\text{PP}}^2}{4} m_{\text{PP}} + \left(l_{\text{PP}}^2 + \frac{l_{\text{MP}}^2}{4}\right) m_{\text{MP}} + \left(l_{\text{PP}}^2 + l_{\text{MP}}^2 + \frac{l_{\text{DP}}^2}{4}\right) m_{\text{DP}}$
p_2	$l_{\text{PP}} l_{\text{MP}} (m_{\text{MP}} + 2m_{\text{DP}})$
p_3	$l_{\text{PP}} l_{\text{DP}} m_{\text{DP}}$
p_4	$l_{\text{MP}} l_{\text{DP}} m_{\text{DP}}$
p_5	$J_{\text{PIP}} + J_{\text{DIP}} + \frac{l_{\text{MP}}^2}{4} m_{\text{MP}} + \left(l_{\text{MP}}^2 + \frac{l_{\text{DP}}^2}{4}\right) m_{\text{DP}}$
p_6	$\frac{1}{2} l_{\text{PP}} l_{\text{MP}} (m_{\text{MP}} + 2m_{\text{DP}})$
p_7	$\frac{1}{2} l_{\text{PP}} l_{\text{DP}} m_{\text{DP}}$
p_8	$J_{\text{DIP}} + \frac{l_{\text{DP}}^2}{4} m_{\text{DP}}$
p_9	$\frac{1}{2} l_{\text{MP}} l_{\text{DP}} m_{\text{DP}}$

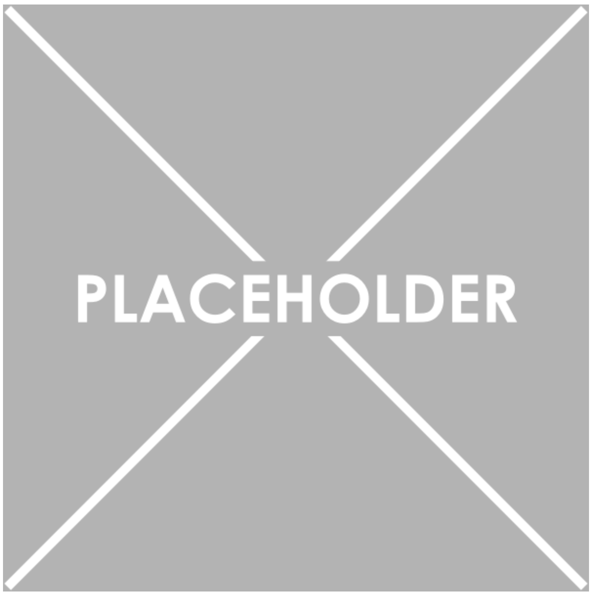

Figure 3.4: Circuit diagram of a brushed DC motor.

3.2 Brushed DC Motor model

In our system we're using a 12 V Brushed DC motor from Pololu [**Cite to motor**]. I've based my dynamical model from [**Cite to article describing Brushed DC Motor**]. The circuit diagram of a brushed DC motor is shown in Section 3.2. Equation (3.16) describes the mechanical dynamics of the motor, and Equation (3.17) describes the electrical dynamics of the motor, defined as:

$$J_m \frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + B_m \frac{\mathrm{d}\theta}{\mathrm{d}t} = \tau_m - \tau_l \tag{3.16}$$

$$R_a I_a + L_a \frac{\mathrm{d}I_a}{\mathrm{d}t} + E_b = E_a. \tag{3.17}$$

Here, θ defines the angle of the motor. The torque generated by the motor is given by $\tau_m = K_t I_a$, where K_t is the torque constant of the motor. The load torque τ_l represents the torque exerted by the finger on the motor through the cable, as well as friction between the motor and the finger. The friction within the rotor of the motor is modeled with $B_m \frac{\mathrm{d}\theta}{\mathrm{d}t}$, where B_m is a constant friction coefficient. The inertia of the rotor is given by J_m. The back electromotive force (EMF) is given by $E_b = K_b \frac{\mathrm{d}\theta}{\mathrm{d}t}$, where K_b is the back EMF constant of the motor. The armature voltage E_a is the control input to the system. R_a and L_a are the armature resistance and inductance, respectively. With the change to τ_m and E_b, we can rewrite the equations as:

$$J_m \frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + B_m \frac{\mathrm{d}\theta}{\mathrm{d}t} = K_t I_a - \tau_l \tag{3.18}$$

$$R_a I_a + L_a \frac{\mathrm{d}I_a}{\mathrm{d}t} + K_b \frac{\mathrm{d}\theta}{\mathrm{d}t} = E_a. \tag{3.19}$$

3.2.1 Estimating motor parameters

To estimate the parameters of the motor, we will use the datasheet from the manufacturer [**Cite motor datasheet**]. Relevant information extracted from the datasheet is shown in Table 3.2. Considering the motor at stall, we have $I_a = I_{a,stall}$ and $\tau_l = \tau_{l,stall}$. In addition we have $\frac{\mathrm{d}\theta}{\mathrm{d}t} \equiv 0$, and at stationarity, $\frac{\mathrm{d}I_a}{\mathrm{d}t} \equiv 0$. This allows us to rewrite Equation (3.19) as:

$$R_a I_{a,stall} = E_a \implies R_a = \frac{E_a}{I_{a,stall}} \approx 2.44\,\Omega. \tag{3.20}$$

Further, when the motor is stalled, we should expect the torque generated by the motor to equal the stall torque, giving us:

$$K_t I_{a,stall} = \tau_{l,stall} \implies K_t = \frac{\tau_{l,stall}}{I_{a,stall}} \approx 0.44\,\mathrm{N\,m\,A^{-1}}. \tag{3.21}$$

For the case when the motor is running at no-load speed, we have $I_a = I_{a,nl}$ and $\frac{\mathrm{d}\theta}{\mathrm{d}t} = \omega_{nl}$, and at stationarity, $\frac{\mathrm{d}I_a}{\mathrm{d}t} \equiv 0$. This allows us to rewrite Equation (3.19) as:

$$R_a I_{a,nl} + K_b \omega_{nl} = E_a \implies K_b = \frac{E_a - R_a I_{a,nl}}{\omega_{nl}} \approx 0.845\,\mathrm{V\,s\,rad^{-1}}. \tag{3.22}$$

Further, at no-load speed, we have $\tau_l \equiv 0$, and $\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} \equiv 0$. This allows us to rewrite Equation (3.18) as:

$$B_m \omega_{nl} = K_t I_{a,nl} \implies B_m = \frac{K_t I_{a,nl}}{\omega_{nl}} \approx 6.47\,\mathrm{mN\,m\,s\,rad^{-1}}. \tag{3.23}$$

Sadly it's not possible to estimate the inertia of the rotor J_m, nor the armature

Table 3.2: Relevant motor parameters extracted from the datasheet of the motor.

Parameter	Symbol	Value
Armature Voltage	E_a	12 V
Stall Torque	$\tau_{l,stall}$	2.158 N m
Stall Current	$I_{a,stall}$	4.9 A
No-load Speed	ω_{nl}	13.61 rad/sec
No-load Current	$I_{a,nl}$	0.20 A

inductance L_a from the datasheet. For simulation purposes, we will define $J_m = 0.093\,\mathrm{kgm}^2$ and $L_a = 6\,\mathrm{mH}$, which were given as common values brushed DC motors in [**Cite to article describing Brushed DC Motor**]. As for reasons that come later, estimating J_m offline is not necessary for our control design, but estimating L_a is. There are many ways to do this. One way that's probably a bit overkill, but that I think will be the most convenient, is to use a parameter identification method.

3.2.2 Estimating the armature inductance

To estimate the armature inductance, we can use a parameter identification method. We'll start by converting Equation (3.19) to the Laplace domain, giving us:

$$E_a(s) = (L_a s + R_a)I_a(s) + K_b s\theta(s). \tag{3.24}$$

Since we only measure $I_a(t)$ and not its derivative, we have to apply a proper filter of order 1, defined as:

$$\Lambda_{I_a}(s) = s + \lambda_0, \quad \lambda_0 > 0. \tag{3.25}$$

Applying the filter to Equation (3.24) gives us:

$$\frac{1}{\Lambda_{I_a}(s)}E_a(s) = \frac{1}{\Lambda_{I_a}(s)}(L_a s + R_a)I_a(s) + \frac{1}{\Lambda_{I_a}(s)}K_b s\theta(s).$$

Ordering it around, moving all measured signals and known parameters to the left, and all factors with the unkown parameter L_a to the right, we get:

$$\underbrace{\frac{1}{\Lambda_{I_a}(s)}[E_a(s)] - \frac{1}{\Lambda_{I_a}(s)}[R_a I_a(s)] - \frac{s}{\Lambda_{I_a}(s)}[K_b\theta(s)]}_{z} = \underbrace{\frac{s}{\Lambda_{I_a}(s)}[I_a(s)]}_{\Phi}\,\underbrace{L_a}_{\theta^*} \tag{3.26}$$

$$\implies z = \theta^*\Phi.$$

Hence we've transformed the system to a SPM like discussed in [**Add reference to Theory:SPM**]. Using the gradient method developed in [**Add reference to Theory:**

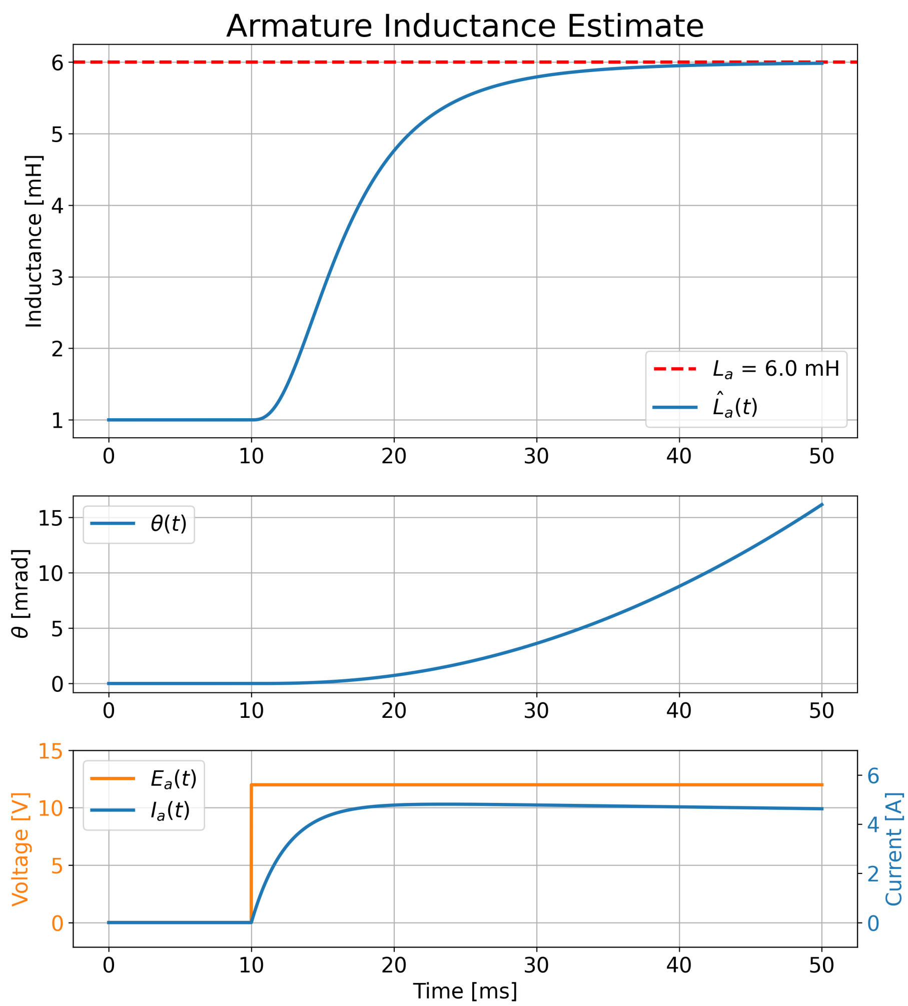

Figure 3.5: Parameter identification of the armature inductance L_a using a step input in the armature voltage E_a.

Gradient method], with $\lambda_0 = 10$, $\Gamma = 10$, and $\theta_0 = \hat{L}_{a0} = 1\,\text{mH}$, while applying a step input $E_a = 12\,\text{V}$ at $t = 10\,\text{ms}$, the resulting estimate of $\hat{L}_a(t)$ is shown in Section 3.2.2. It is clear that the estimate converges exponentially fast to the true value. Since this is a static parameter that only needs to be estimated once, L_a will be assumed known for the further controller design.

3.2.3 Mass-spring-damper load

In [**Reference to finger model**] we assumed that the finger could be modelled as several torsional mass-spring-damper systems. Since we can't measure the finger-angles, the full model isn't observable, and can't be used for control design. Instead, we'll assume that finger can be modelled as a single mass-spring-damper system, acting on the motor as τ_l. Hence we'll model the load torque τ_l as:

$$\tau_l = J\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + c\frac{\mathrm{d}\theta}{\mathrm{d}t} + k(\theta - \theta_{eq}), \tag{3.27}$$

where J is the total inertia of the finger, c is the total damping coefficient of the finger, k is the total stiffness of the finger, and θ_{eq} is the equilibrium angle of the finger. Substituting this into Equation (3.18), we get the updated dynamics:

$$(J_m + J)\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + (B_m + c)\frac{\mathrm{d}\theta}{\mathrm{d}t} + k\theta = K_t I_a + k\theta_{eq} \tag{3.28}$$

$$R_a I_a + L_a\frac{\mathrm{d}I_a}{\mathrm{d}t} + K_b\frac{\mathrm{d}\theta}{\mathrm{d}t} = E_a. \tag{3.29}$$

3.2.4 State-space model of brushed DC motor with load

To design a controller for the system, we need to express the dynamics in state-space representation, shown in Section 2.3.1. Defining $\omega \triangleq \frac{\mathrm{d}\theta}{\mathrm{d}t}$, we can rewrite the Equation (3.28) and Equation (3.29) as:

$$\begin{aligned}
\frac{\mathrm{d}\omega}{\mathrm{d}t} &= \frac{1}{J_m + J}\left(K_t I_a - (B_m + c)\omega - k(\theta - \theta_{eq})\right) \\
\frac{\mathrm{d}I_a}{\mathrm{d}t} &= \frac{1}{L_a}(E_a - R_a I_a - K_b\omega).
\end{aligned}$$

Written more compactly, we have:

$$\underbrace{\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix}\theta \\ \omega \\ I_a\end{bmatrix}}_{\frac{d}{dt}\mathbf{x}} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ -\frac{k}{J_m+J} & -\frac{B_m+c}{J_m+J} & \frac{K_t}{J_m+J} \\ 0 & -\frac{K_b}{L_a} & -\frac{R_a}{L_a}\end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix}\theta \\ \omega \\ I_a\end{bmatrix}}_{\mathbf{x}} + \underbrace{\begin{bmatrix}0 \\ 0 \\ \frac{1}{L_a}\end{bmatrix}}_{\mathbf{B}} \underbrace{E_a}_{\mathbf{u}} + \underbrace{\begin{bmatrix}0 \\ \frac{k\theta_{eq}}{J_m+J} \\ 0\end{bmatrix}}_{\mathbf{d}} \tag{3.30}$$

$$\implies \frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{d}.$$

In addition we have an encoder measuring θ, aswell as an ammeter measuring I_a. Hence the output of the system is given by:

$$\underbrace{\begin{bmatrix}\theta \\ I_a\end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix}1 & 0 & 0 \\ 0 & 0 & 1\end{bmatrix}}_{\mathbf{C}} \underbrace{\begin{bmatrix}\theta \\ \omega \\ I_a\end{bmatrix}}_{\mathbf{x}} \tag{3.31}$$

$$\implies \mathbf{y} = \mathbf{C}\mathbf{x}.$$

Chapter 4

Adaptive Controller

4.1 Controllability of the system

Based on the state-space representation our system in Section 3.2.4, we can check if the system is controllable, as described in Section 2.3.2. The controllability matrix $\mathcal{C}$ for our system is given by:

$$\mathcal{C} = [\mathbf{B}, \mathbf{AB}, \mathbf{A}^2\mathbf{B}]. \tag{4.1}$$

Substituting for our specific $\mathbf{A}$ and $\mathbf{B}$, we get:

$$\mathcal{C} = \begin{bmatrix} 0 & 0 & \frac{K_t}{L_a(J_m+J)} \\ 0 & \frac{K_t}{L_a(J_m+J)} & -\frac{K_t[L_a(B_m+c)+R_a(J_m+J)]}{L_a^2(J_m+J)^2} \\ \frac{1}{L_a} & -\frac{R_a}{L_a^2} & -\frac{K_bK_tL_a+R_a^2(J_m+J)}{L_a^3(J_m+J)} \end{bmatrix}. \tag{4.2}$$

Clearly, $\text{rank}(\mathcal{C}) = 3$, which is equal to the number of states. Therefore, the system is controllable.

4.2 MRAC implementation

Bibliography

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